IO Checkout Tool

Setup & User Guide

Industrial I/O Commissioning Application

v2.9

For: Field Technicians & System Administrators

Platform: Windows Tablet / Desktop

Last Updated: March 2026

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1. Installation

The IO Checkout Tool runs as a self-contained web application on a Windows machine. It can be deployed in two ways:

Method	Best For	Auto-Start	Service
Portable Folder	Quick deployment, testing, temporary use	No	No
Windows Installer	Permanent installation, production use	Yes (on boot)	Yes

1.1 First Startup

Regardless of installation method, the first startup involves:

1. Application starts on **port 3000** (HTTP) and **port 3002** (WebSocket)
2. A blank SQLite database is created automatically
3. A default `config.json` file is generated
4. The default admin account is ready with PIN `111111`

> **Important:**
> Change the default admin PIN immediately after first login. Anyone with the PIN can access the system and modify settings.

1.2 Firewall Setup

Two ports are required for other devices (tablets, laptops) to connect:

Port	Protocol	Purpose

`3000`	TCP	Web UI — technicians access this in their browser
`3002`	TCP	WebSocket — real-time PLC state updates

> **Automatic:**
> Firewall rules are configured automatically when `START.bat` runs. No manual firewall setup is needed.

> **Note:**
> Port 3102 is used internally (localhost only) and is not exposed.

1.3 Portable Folder Usage

The portable distribution is a single folder containing everything needed — no installation required.

Starting the Application

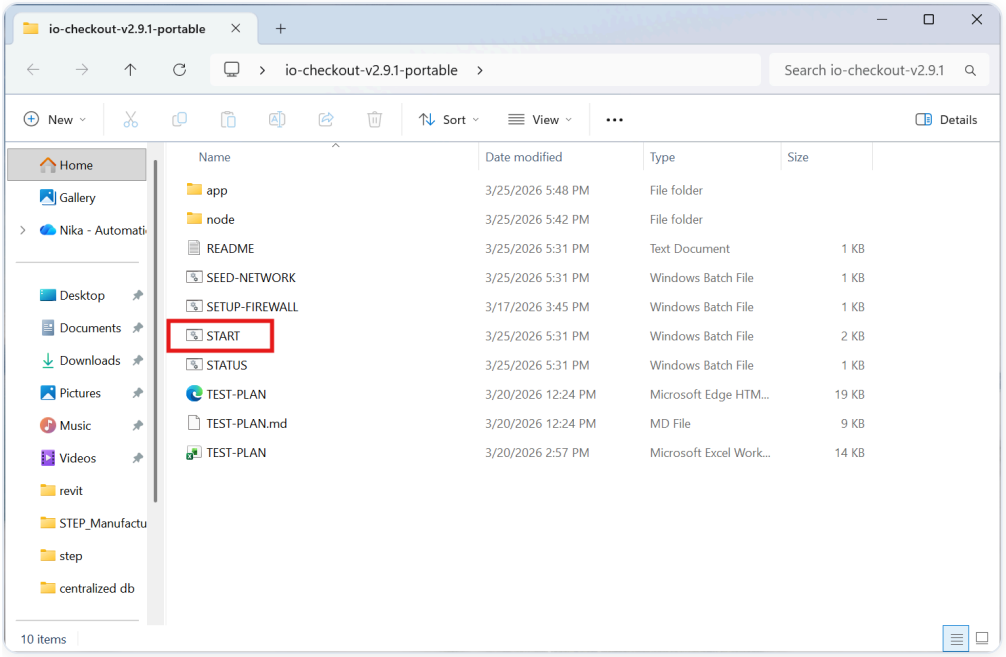
1. Double-click `START.bat` in the portable folder. A command window will open showing startup logs.
2. Wait for the message *"All servers started successfully"* (takes ~5 seconds).
3. Open a browser and navigate to `http://localhost:3000` (or `http://SERVER_IP:3000` from another device).

Stopping the Application

Close the command window that was opened by `START.bat`, or run `STOP.bat`.

Checking Status

Run `STATUS.bat` to see if the application is running and which IP addresses are available for other devices to connect.



Folder Structure

```
portable/
  START.bat              ← Start the application (auto-opens firewall)
  STOP.bat               ← Stop the application
  STATUS.bat             ← Check if running + show IPs
  SETUP-FIREWALL.bat     ← Firewall rules (called by START.bat automatically)
  config.json            ← Configuration (auto-created)
  database.db            ← SQLite database (auto-created)
  app/                   ← Application files (do not modify)
  node/                  ← Bundled Node.js runtime
```

1.4 Windows Installer (Service)

The installer sets up IO Checkout as a **Windows service** that starts automatically on boot and restarts on crash.

Installation

1. Run the installer `.exe` as Administrator.
2. Follow the setup wizard — default install path is `C:\Program Files\IOCheckout\`.

3. The service starts automatically after installation. Access at `http://localhost:3000`.

Data Locations (Preserved Across Upgrades)

Item	Location
Database	`C:\ProgramData\IOCheckout\database.db`
Configuration	`C:\ProgramData\IOCheckout\config.json`
Logs	`C:\ProgramData\IOCheckout\logs\`

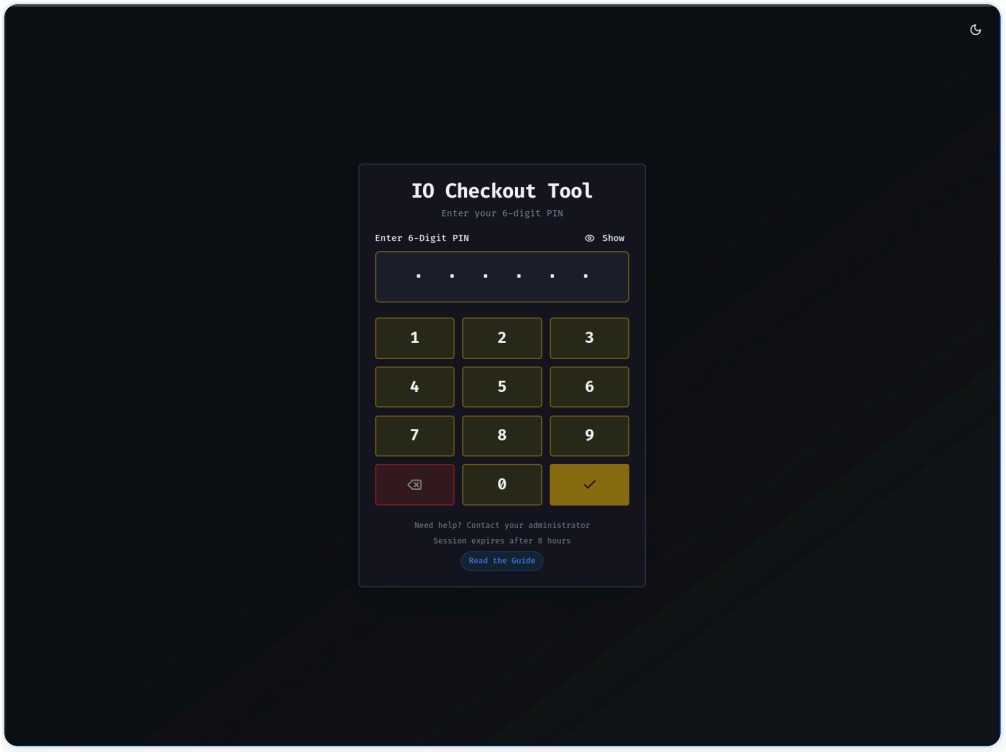
Service Management

```
# Check service status
nssm status IOCheckout

# Restart the service
nssm restart IOCheckout

# Stop the service
nssm stop IOCheckout

# Or use Windows Services: services.msc → "IOCheckout"
```

2. Logging In

The application uses **PIN-based authentication**. No usernames — just a 6-digit numeric PIN.

1. Open the application in a browser: `http://SERVER_IP:3000`
2. Enter your 6-digit PIN on the login screen.
3. Press **Enter** or tap the login button.



> **Default Admin PIN:**
> `111111` — Change this immediately in *Manage Users* after first login.

Sessions last **8 hours**. After that, you'll need to re-enter your PIN.

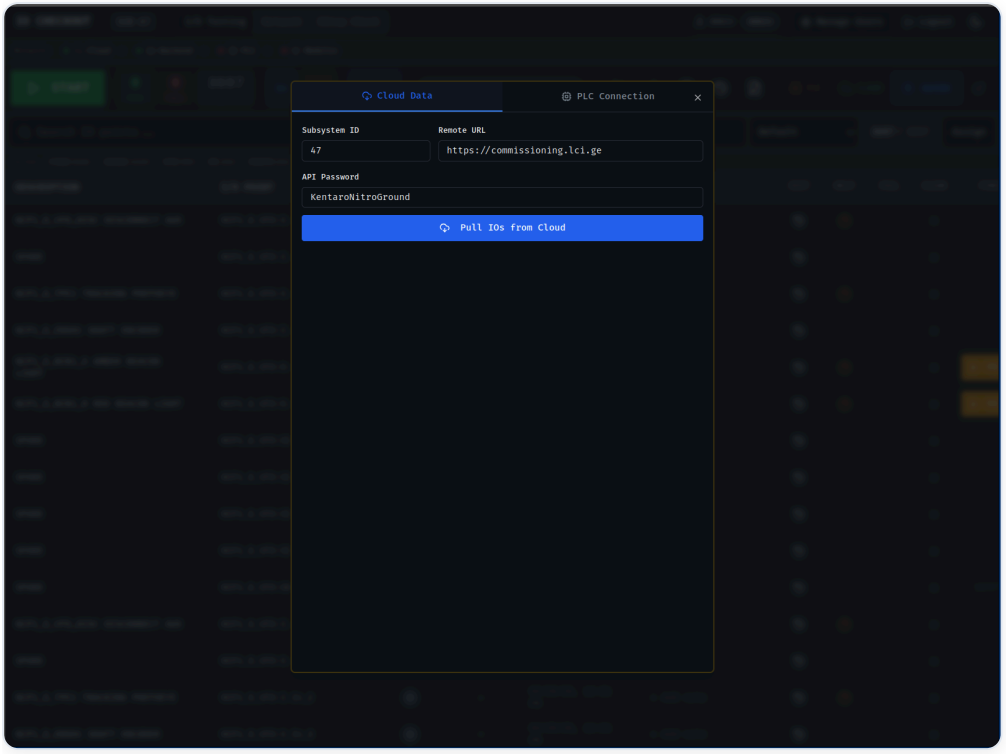
3. Initial Admin Setup

After first login with the default admin PIN, complete these steps to configure the system:

3.1 Cloud Configuration

The cloud connection allows syncing IO definitions and test results with the central commissioning server.

1. Click the **gear icon** in the toolbar to open Configuration.
2. Enter the **Cloud URL** (e.g., `https://commissioning.lci.ge`).
3. Enter the **Subsystem ID** — this identifies which subsystem (MCM) this tool manages.
4. Enter the **API Password** (project API key from the cloud app).
5. Click **Save**. The cloud status indicator in the breadcrumb should turn green.

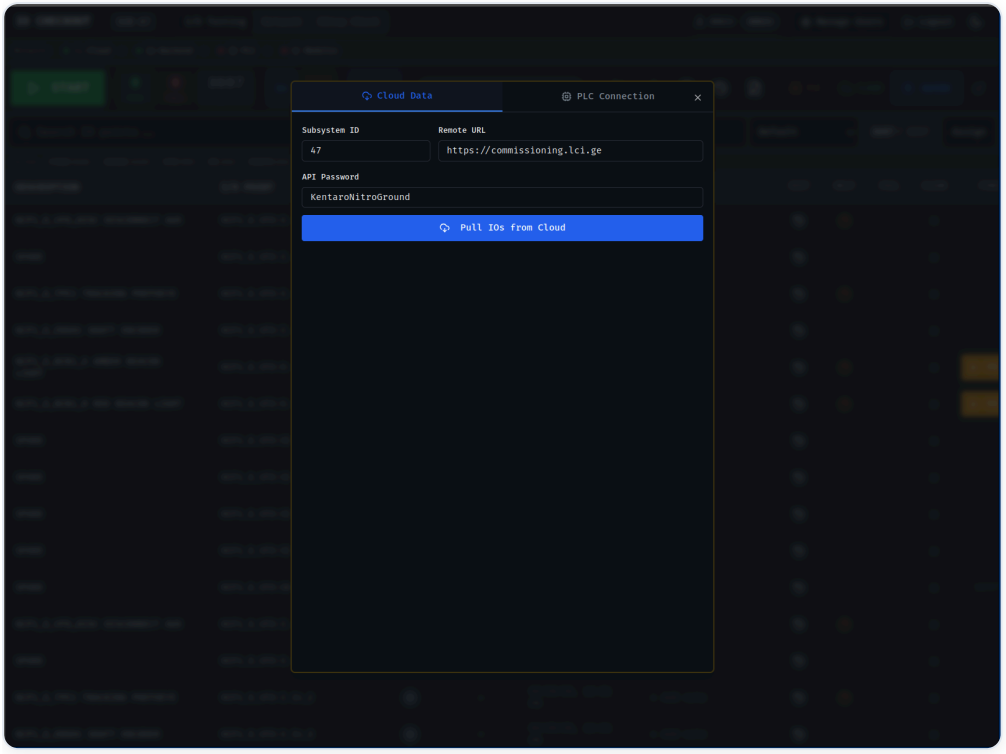


3.2 Pulling IOs from Cloud

After configuring the cloud connection, pull the IO definitions:

1. Click **"Pull IOs"** in the toolbar (admin only).
2. Wait for the download to complete. The IO grid will populate with all IO points for your subsystem.

> **Warning:**
> Pulling IOs replaces ALL existing IO definitions. A backup of the current database is created automatically before the pull.



3.3 PLC Configuration & Connection

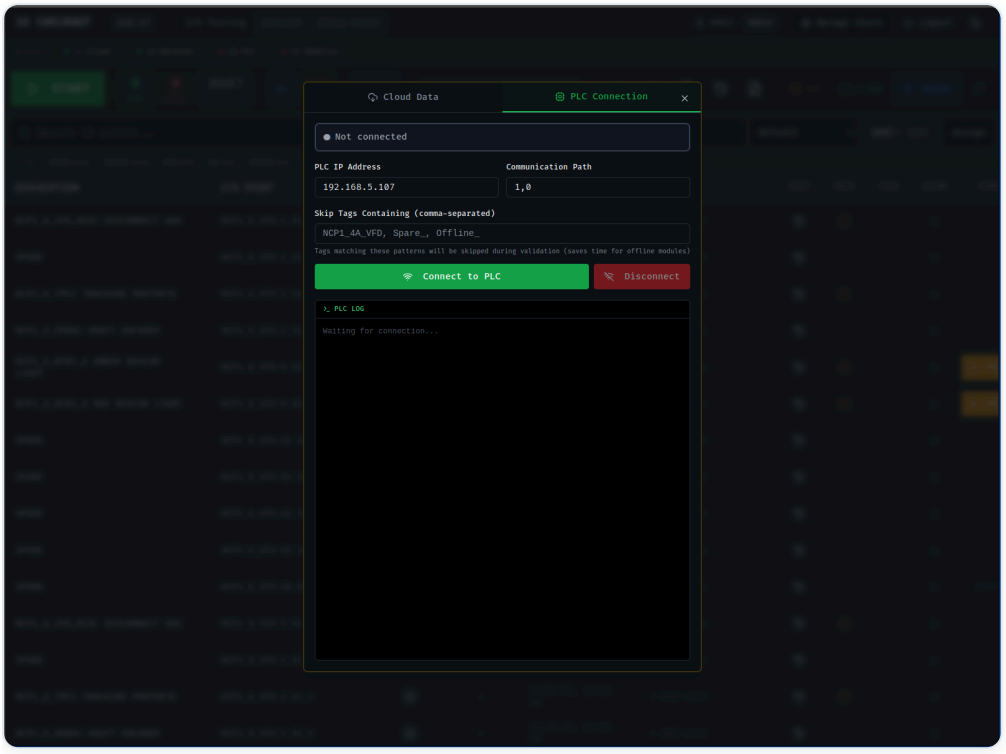
Configure the PLC connection to enable live tag reading:

1. Open **Configuration** (gear icon).
2. Enter the **PLC IP Address** (e.g., `11.200.1.1`).
3. Set the **Communication Path** — usually `1,0` for standard Allen-Bradley Ethernet/IP.
4. Click **Save**, then click the **Connect** button (plug icon) in the toolbar.

Connection states:

Indicator	Meaning
● Green	Connected — PLC communication active, tags reading every 75ms

● Amber	Reconnecting — PLC was lost, auto-reconnecting every 5 seconds
● Gray	Disconnected — Not connected to PLC



> **Auto-Reconnect:**
> If the PLC loses power or the network drops, the tool automatically reconnects every 5 seconds. No admin intervention needed — testing resumes automatically.

3.4 Managing Users

Create PIN accounts for each technician:

1. Click **"Users"** (people icon) in the toolbar.
2. Click **"Add User"**.
3. Enter the technician's **name** and a unique **6-digit PIN**.

4. Set role: **Admin** (full access) or **User** (testing only, no config/PLC/cloud).

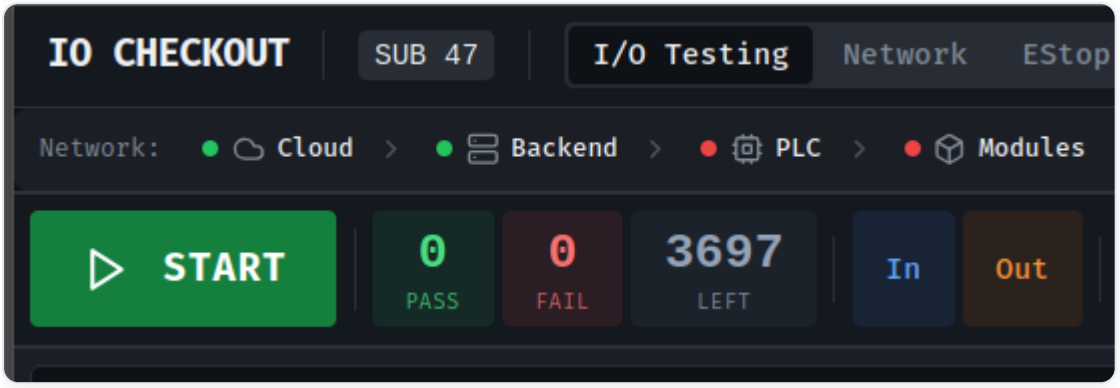
Screenshot: Manage Users dialog (add users with Name, PIN, and Role)

4. Testing Workflow

4.1 Starting & Stopping Testing

Testing mode must be started before IO states are monitored:

1. Click the **START** button in the toolbar. The app begins reading PLC tags every 75ms.
2. IO states appear in real-time in the grid. `FALSE` states show in gray, `TRUE` in green.
3. Click **STOP** to pause testing. Tag reading stops and states freeze.

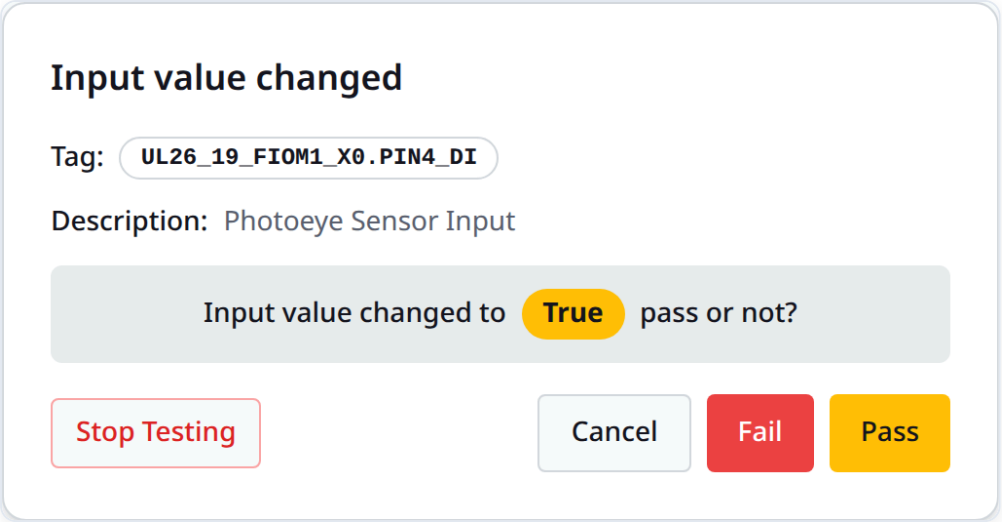


4.2 Testing Inputs

Input testing is **automatic** — the tool detects state changes from the PLC:

1. Go to the physical panel and trigger the device (press button, activate sensor, etc.).
2. The IO state changes from `FALSE` to `TRUE` in the app.
3. A **Pass/Fail dialog** automatically appears asking for the test result.

4. Tap **Pass** (green) or **Fail** (red). Optionally add a comment.



> **Tip:**
> The dialog shows which IO changed and its current PLC state. You can search and filter the IO grid using the search bar at the top.

4.3 Testing Outputs (Fire)

Output testing requires **manually triggering** the output from the app:

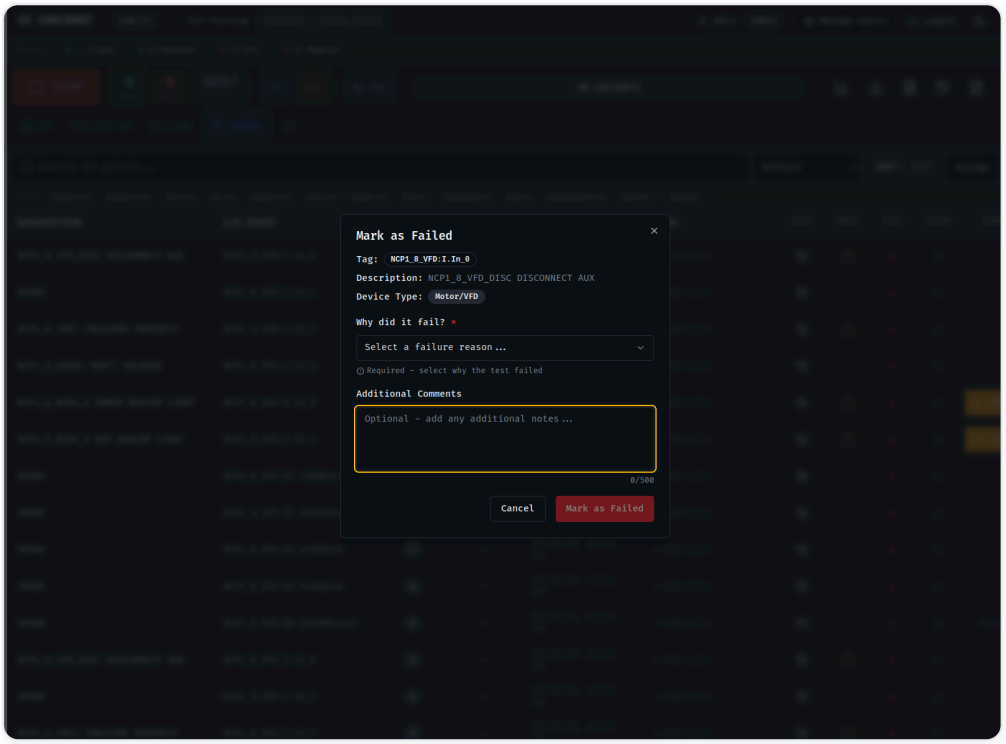
1. Find the output IO in the grid.
2. Click the **fire icon** (lightning bolt) to send a write command to the PLC.
3. Verify the physical device activates (light turns on, motor runs, etc.).
4. Mark as **Pass** or **Fail**.

> **Safety:**
> Only fire outputs when it is safe to do so. Ensure the area is clear and personnel are aware before activating outputs.

4.4 Handling Failures

When an IO fails testing:

1. Select **Fail** in the Pass/Fail dialog.
2. Choose a **failure mode** from the dropdown (e.g., "No Response", "Wrong Device", "Wiring Issue").
3. Add a **comment** describing the issue.
4. The IO is marked red in the grid with the failure details.



4.5 Adding Comments

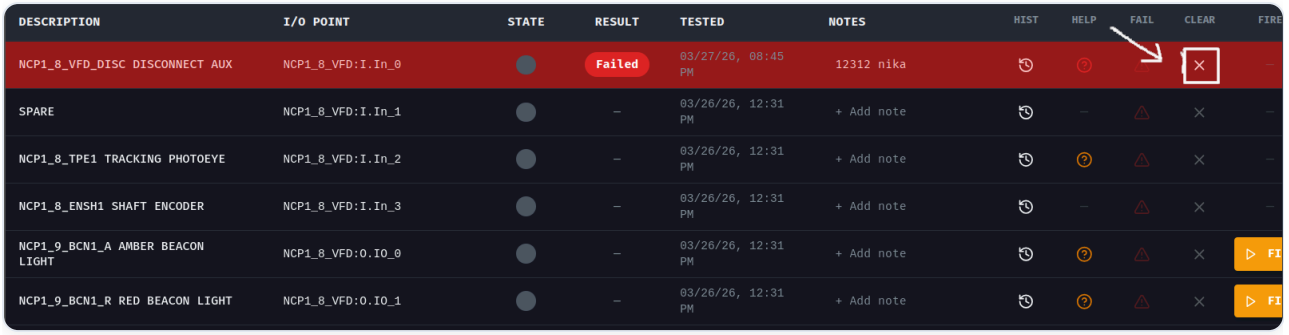
Comments can be added to any IO at any time — click the IO row and type in the comment field. Comments sync to the cloud automatically.

4.6 Resetting Results

To re-test an IO that was already marked Pass or Fail:

1. Click the IO row.

2. Click **Clear/Reset** to remove the existing result.
3. The IO returns to "Not Tested" state and can be tested again.



> **Note:**
> Clearing a result does NOT delete the test history. The previous result remains in the audit trail.

5. Cloud Sync

The tool syncs test results with the cloud automatically:

Direction	Timing	What Syncs
Local → Cloud	Instant (1-2 seconds)	Every Pass, Fail, Comment, Reset
Cloud → Local	Via SSE (real-time)	Results from other instances, admin changes

Offline Resilience

If the cloud connection is lost:

- Testing continues normally — all results save locally.
- Pending changes queue in the database.
- When connectivity returns, all queued results sync automatically.

Multi-Instance Support

Multiple local tool instances can test the same subsystem simultaneously. Results merge automatically — the cloud is the source of truth for shared data.

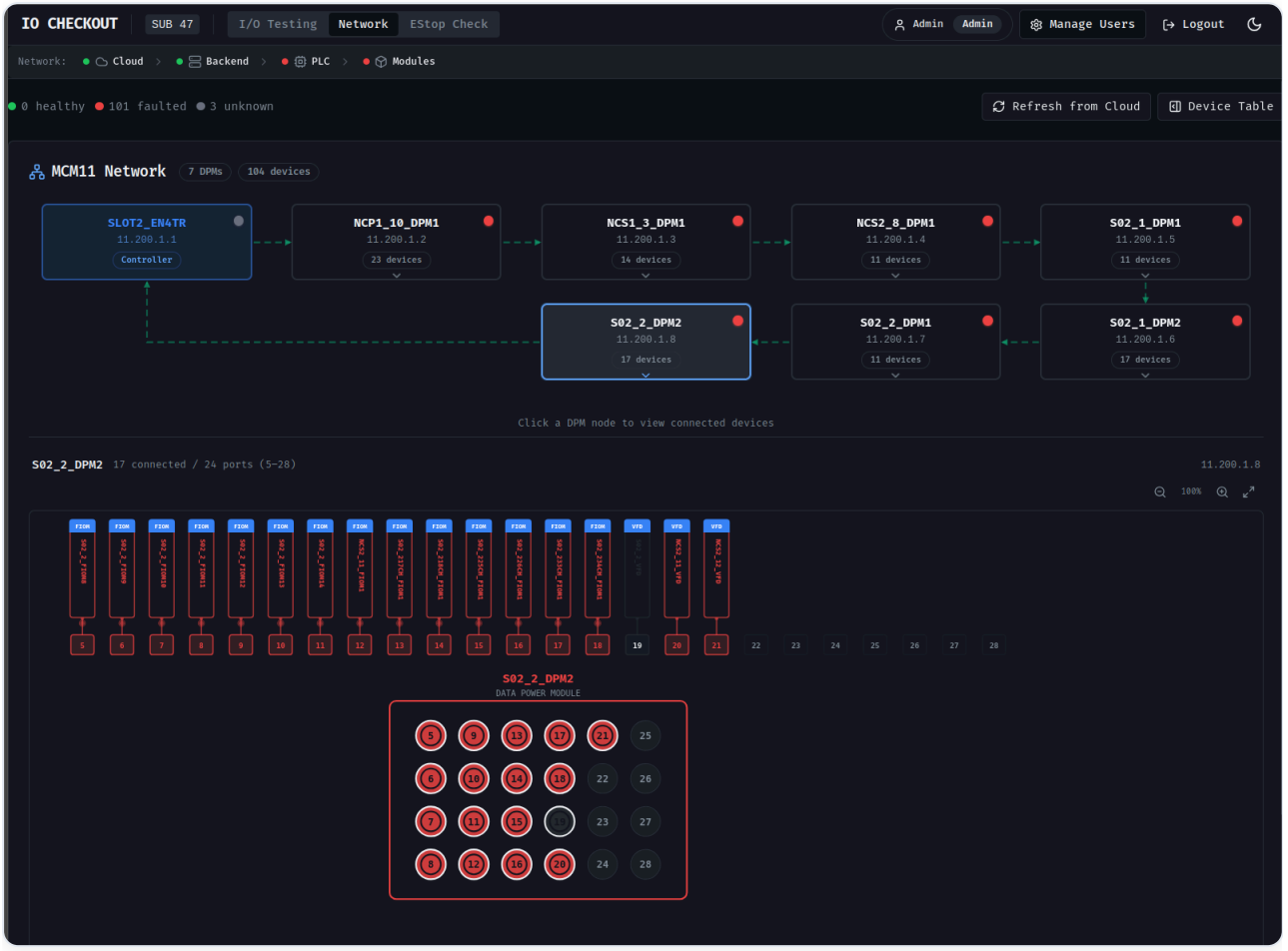
6. Network Topology

The Network tab displays the network devices connected to each DPM (Data Power Module) and their live connection status from the PLC.

What It Shows

- **DPM ring layout** — MCM controller and DPM nodes arranged in a serpentine pattern
- **Star diagram** — Click a DPM to see all network devices connected to its ports (VFDs, FIOMs, PMMs, etc.)
- **FIOM sub-devices** — Click a FIOM device to see devices connected to its X-ports
- **Device connection status** — Each device shows live `ConnectionFaulted` status from the PLC:
  - ● Green — device is communicating normally
  - ● Red — device connection is faulted
  - ● Gray — no PLC data (disconnected or not monitoring)

> **Note:**
> The network view monitors individual device connection status (ConnectionFaulted tags). It does not monitor DLR ring status — only whether each device on the DPM ports is communicating or faulted.


IO CHECKOUT
SUB 47
I/O Testing
Network
EStop Check
Admin
Admin
Manage Users
Logout
Network:
Cloud
Backend
PLC
Modules
0 healthy
101 faulted
3 unknown
Refresh from Cloud
Device Table
MCM11 Network
7 DPMs
104 devices
SLOT2_EN4TR
11.200.1.1
Controller
NCP1_10_DPM1
11.200.1.2
23 devices
NCS1_3_DPM1
11.200.1.3
14 devices
NCS2_8_DPM1
11.200.1.4
11 devices
S02_1_DPM1
11.200.1.5
11 devices
S02_2_DPM2
11.200.1.8
17 devices
S02_2_DPM1
11.200.1.7
11 devices
S02_1_DPM2
11.200.1.6
17 devices
Click a DPM node to view connected devices
S02_2_DPM2 17 connected / 24 ports (5–28)
11.200.1.8
100%
S02_2_DPM2
DATA POWER MODULE

7. Troubleshooting

Problem	Solution
Can't connect from tablet	Restart the app with `START.bat` (re-applies firewall rules). Check `STATUS.bat` for the correct IP.
PLC won't connect	Verify PLC IP and path (`1,0`). Check network cable. Ensure PLC is powered on.
IOs not showing	Pull IOs from cloud first. Check cloud URL and subsystem ID in configuration.
State not changing	Ensure testing mode is started (START button). Check PLC connection indicator.
Results not syncing	Check cloud status indicator. Verify cloud URL and API password.
App won't start	Check if port 3000 is in use: `netstat -ano | findstr :3000`. Kill the process or change port.
"Reconnecting" indicator	PLC lost connection — auto-reconnects every 5s. No action needed.
Forgot admin PIN	Delete `database.db` and restart (resets everything). Or contact system admin.

8. Quick Reference

Ports

Port	Purpose	Firewall
`3000`	Web UI (HTTP)	Open
`3002`	WebSocket (real-time updates)	Open
`3102`	Internal broadcast API	Do NOT open

Default Credentials

Account	PIN	Role
Admin	`111111`	Full access (config, PLC, users, cloud)

Key Files

File	Purpose	Safe to Delete?
`database.db`	All IO data, test results, users	No (resets everything)
`config.json`	PLC IP, cloud URL, subsystem ID	No (lose configuration)
`logs/`	Application logs (auto-rotating)	Yes (recreated automatically)

Toolbar Icons

Icon	Action	Access
Play/Stop	Start/Stop testing mode	All users

Plug	Connect/Disconnect PLC	Admin only
Cloud	Pull IOs from cloud	Admin only
Gear	Configuration (PLC, cloud settings)	Admin only
People	Manage user accounts	Admin only
Download	Export results to CSV	All users